Agri_Empower Training Series

Yam Mastery: Intermediate Level (Commercial Optimization)

Module 1: Production and Management

Scientific Tuber Prep & Spacing

- **Standardized Sett Selection:** Grading seed yams into 500g - 1kg categories for uniform field growth and predictable harvest dates.
- **The Curing Protocol:** 48-hour shade-curing to develop a "suberized" layer (callus), which acts as a biological shield against soil-borne pathogens.
- **Field Layout:** Moving to 1-meter inter-row spacing to facilitate movement and air circulation, reducing humidity-related foliage diseases.
- **NPK Side-Dressing:** Placing fertilizer 10-15cm away from the vine base to ensure the active root zone absorbs nutrients without salt-burn damage.

Special Feature: Field Soil Testing Process

Virtual Training Guide:

1. **Sample Pattern:** Walk the farm in a "Zig-Zag" pattern. Collect samples from 5-10 different spots to get a representative average.
2. **The V-Cut Method:** Clear surface debris. Dig a V-shaped hole 15-20cm deep (the tuber zone). Slice a 2cm thick piece from the side of the V.
3. **Mixing:** Combine all slices in a clean plastic bucket. Remove stones and roots.
4. **The Test:** Take a small scoop of the mixed soil. Add distilled water. Use a pH strip or digital probe.
5. **Interpretation:**
   - **5.5 - 6.5:** Perfect. Nutrients are "unlocked."
   - **Below 5.0:** Acidic. Requires Agricultural Lime.

    - **Above 7.0:** Alkaline. Requires organic manure/compost to lower pH.

Module 2: Pest, Disease, and Soil Health

- **Integrated Weed Management (IWM):** Combining manual weeding with calibrated pre-emergence herbicide use for 1-hectare+ plots.
- **Virus Scouting:** Identifying "Mottle and Vein-clearing" (Yam Mosaic Virus). Immediate removal of infected plants to prevent aphid-spread.
- **Soil Amendment:** Using "Green Manure" (planting legumes in the off-season) to naturally fix nitrogen.

Module 3: Post-Harvest and Storage

- **Therapeutic Curing:** Keeping harvested tubers in a high-humidity environment (80-90%) for 4-7 days to heal harvest scratches.
- **Improved Storage Design:** Constructing "Box-Barns" with raised wooden platforms and "Rat Guards" (conical metal sheets) on supporting poles.
- **Sprout Management:** Regular "De-sprouting" to prevent the tuber from using its stored sugar/energy to grow vines in the barn.

Module 4: Processing and Value Addition

- **Quality Yam Flour (Elubo):** Implementation of parboiling at exactly 60-70°C to prevent starch gelatinization while killing spoilage organisms.
- **Packaging & Branding:** Introduction to HDPE (High-Density Polyethylene) bags to prevent moisture re-absorption.

Module 5: Agribusiness and Markets

- **Cost-Benefit Analysis:** Calculating "ROI" (Return on Investment) by factoring in seed costs, labor, logistics, and storage losses.
- **Collective Logistics:** Using the "Doing Good" cooperative model to pool harvests for lower transport costs to urban centers.

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